Spherical DRA Power Combining & CP — Study Roadmap

Sep 18, 2026 · @mansi

This is a working study document for a BTech project under Prof. Utpal Dey: a millimeter-wave spherical dielectric resonator antenna (DRA) that combines power from multiple feeds and targets circular polarization (CP), building directly on two of his group's MAPCON papers (stripline-fed spherical DRA with assembly features; fundamental modes of a spherical DR at mmWave).

Part A — Fundamentals: Waves, Modes, and Mode Profiles

A guided-wave mode is a self-consistent solution of Maxwell's equations that keeps its transverse field shape while propagating along z with only a phase/amplitude change (∝ e^(−jβz)). Everything below is the minimum needed to answer "what is a TEM/TE/TM mode" confidently and to explain what a "mode profile" is.

Transverse waves

A wave is transverse when its field vectors oscillate perpendicular to the direction of travel. In free space, a plane EM wave is purely transverse: both E and H lie in the plane perpendicular to the propagation direction k, and E ⊥ H ⊥ k. Inside a guiding structure (waveguide, transmission line), this simple picture only survives for TEM modes; TE and TM modes have a longitudinal field component precisely because a boundary (conductor or dielectric interface) forces it.

Mode families

Mode	Ez	Hz	Exists in	Typical example
TEM (transverse electromagnetic)	0	0	Structures with ≥2 separate conductors	Coax, parallel-plate line, ideal stripline
TE (transverse electric, H-mode)	0	≠0	Any hollow/dielectric waveguide	Rectangular waveguide TE10, dielectric rod TE01

Mode	Ez	Hz	Exists in	Typical example
TM (transverse magnetic, E-mode)	≠0	0	Any hollow/dielectric waveguide	Circular waveguide TM01, DR TM101
Hybrid (HE/EH)	≠0	≠0	Open dielectric guides (fibers, dielectric rods, real DRAs)	Optical fiber HE11, most practical DRA modes

Why TEM needs two conductors: a TEM mode's transverse fields are literally the static (Laplace-equation) field pattern between two conductors, riding on a traveling wave — that pattern only exists when a return conductor is present to terminate field lines. A single hollow pipe (rectangular/circular waveguide) or a single dielectric rod cannot support TEM; the boundary conditions force at least one longitudinal component, giving TE or TM.

Microstrip and stripline are only quasi-TEM in the strict sense (stripline is very close to true TEM because it is symmetric and fully enclosed between two ground planes; microstrip has a small longitudinal component because fields partly live in air, partly in substrate).

Cutoff frequency and dominant mode

Every TE/TM mode in a closed waveguide has a cutoff frequency f_c below which it cannot propagate (it decays evanescently instead). The dominant mode is the one with the lowest f_c — for rectangular waveguide this is TE10, which is why WR-28 (used in Paper 2) is described as operating in its fundamental TE10 mode. Above f_c, $\beta = k_0\sqrt{(1 - (f_c/f)^2)}$ is real and the mode propagates; below it, β is imaginary and the field only reaches a short distance into the guide (evanescent, as noted for the TM11 mode discussion in Paper 2, whose cutoff of 47.19 GHz is above the WR-28 test band).

Mode profile — what it actually means

A "mode profile" is the transverse field pattern (magnitude and direction of E and/or H) across the guide's cross-section, at a fixed z. It is fixed by the geometry and boundary conditions, not by the source — the same waveguide always produces the same TE10 profile (one half-sine E lobe across the wide dimension) regardless of what feeds it. For dielectric resonators the equivalent concept is the internal/external field pattern of a Mie mode (e.g. the circumferential H-field / radial-ish E-field pattern of TE1q1 shown in Fig. 4a of Paper 2).

Which profiles you'll actually deal with in this project

- Feedlines: quasi-TEM (microstrip/stripline/coax feed network).
- Waveguide characterization (if you replicate Paper 2's method): TE10 in WR-28.
- The resonator itself: spherical dielectric resonator natural modes, classified using Mie theory as TE_nqr / TM_nqr (n = mode order, q = azimuthal index, r = radial index) — not the same TE/TM as in a hollow metal waveguide, but the same underlying logic (which field component is purely transverse). Your project is explicitly scoped to the TE1q1 (magnetic dipole) family — see Part B.

Half-sentence supervisor answer to memorize: "A transverse wave has its fields perpendicular to travel direction; TEM needs two conductors and has no longitudinal field at all; TE/TM each have one longitudinal component (Hz or Ez) forced by a single-conductor or dielectric boundary; the mode profile is the fixed transverse field shape set by geometry, and in this project the profile that matters is the sphere's TE1q1 magnetic-dipole mode."

Part B — Resonators and the Spherical Dielectric Resonator

General resonator theory

A resonator stores energy in oscillating E and H fields that satisfy the boundary conditions at a discrete set of frequencies (like standing waves on a string). Key figure of merit is the quality factor:

$Q = 2\pi \times$ (energy stored) / (energy dissipated per cycle) $= \omega_0 \times$ (stored energy) / (power lost)

Loss mechanisms combine as $1/Q_{total} = 1/Q_{rad} + 1/Q_{diel} + 1/Q_{cond}$ (radiation, dielectric loss $\tan\delta$, conductor loss). A higher-order mode generally has higher Q (less field at lossy boundaries relative to stored energy) — this is exactly the trend Paper 1 cites: "Q-factor of these natural modes increases with the order of the mode." Loaded Q (with the feed/external circuit attached) is always ⩽ unloaded Q; critical coupling is when the external Q equals the internal Q, giving maximum power transfer.

Why dielectric, why sphere

Dielectric resonators avoid conductor (Ohmic) loss entirely — their Q is set by $\tan\delta$ of the material, which for good ceramics (alumina, $\tan\delta \approx 0.005$) beats copper at mmWave. Among DR shapes (rectangular, cylindrical, hemispherical, spherical), the sphere is the

only one with a single tolerance dimension (diameter only — no orientation-dependent tolerance), the smallest planar footprint, and off-the-shelf availability in tight tolerances (±2.5 μm per Paper 2) from ceramic suppliers — which is the whole justification for using it at mmWave where cylindrical/rectangular DR tolerances become unmanageable.

Mie theory in one page

A plane wave hitting a dielectric sphere sets up an infinite series of resonant modes, indexed by order n (n=1 dipole, n=2 quadrupole, …). Mie theory gives four coefficients per order: a_n, b_n (external TM, external TE — what radiates/scatters outward) and c_n, d_n (internal TM, internal TE — fields inside the sphere). The size parameter $x = 2\pi r/\lambda$ (r = sphere radius, λ = free-space wavelength) governs which mode dominates:

- $|b_1| = 1$ at $x \approx 0.95$ → magnetic dipole (TE1q1) resonance.
- $|a_1| = 1$ at $x \approx 1.25$ → electric dipole (TM1q1) resonance.

For a 3 mm alumina sphere ($\varepsilon r \approx 10$), this maps to roughly 28–40 GHz — squarely Ka-band, which is why both papers work there.

TE1q1 vs TM1q1 — the choice that matters for your project

	TE1q1 (magnetic dipole)	TM1q1 (electric dipole)
Dominant when	$\lvert\varepsilon r/\varepsilon m\rvert > 1$ (sphere denser than surroundings — the normal case)	$\lvert\varepsilon r/\varepsilon m\rvert < 1$ (sphere less dense, e.g. air bubble in liquid)
Field inside sphere	Near-linear H-field, circumferential (non-radial) E-field	Radial-ish E-field character
Dielectric filling factor W_1/W	Lower	Higher (more energy actually inside the lossy sphere)
Consequence	Lower dielectric loss → higher Q	Higher dielectric loss → lower Q
Measured Q (Ka-band prototype, Paper 2)	≈2352	≈1363

	TE1q1 (magnetic dipole)	TM1q1 (electric dipole)
Best suited for	On-chip resonators, narrowband filters, sensors (needs high Q)	Antennas needing low Q / wide bandwidth, wideband transitions

Since your project is scoped to **TE**, you inherit the high-Q, narrower-bandwidth character of TE1q1 — good for a well-defined combining/radiating mode, but expect a naturally modest impedance bandwidth (Paper 1's TM-mode antenna already only got ~3% at −10 dB; TE will likely be similar or tighter unless the feed network is designed to broaden it).

External vs internal modes, and where energy actually sits

Internal modes (c_1, d_1) set the field pattern *inside* the sphere; external modes (a_1, b_1) set the *scattered/radiated* field outside it. Total energy $W = W_1$ (inside sphere) + W_2 (reactive, outside) + W_3 (scattered to far field). Equation (3) in Paper 2 relates the internal energy ratio between the two dipole modes to the sphere's complex refractive index — this is the formal basis for the Q-factor comparison above.

How mode resonance is experimentally identified

- Waveguide transmission method (Paper 2): sphere placed inside WR-28; TE1q1 gives a transmission-zero/reflection-zero (perfect coupling) around 28.5 GHz; TM1q1 gives a transmission-zero around 33.1 GHz because it converts the incident TE10 into an evanescent TM11 mode.
- Near-field E/H probing at the resonance frequency, useful at THz where waveguide methods become impractical.
- On-chip parallel-plate test (Paper 2, Fig. 5): sphere sandwiched between two PEC plates with a PMC symmetry plane at the midline — this cancels the TEM radiation mode and isolates the sphere's own Q, feeding via microstrip fringing fields; this is the closest analog to how you'll set up a CST eigenmode study before adding real feeds.

Part C — The Two Reference Papers, and the Gap They Leave Open

Paper 1 — Stripline-fed spherical DRA with assembly features (MAPCON 2023)

- Stack: microstrip feed → stripline (via-fenced) → rectangular iris → supportive layer with an air-hole for sphere self-alignment → metal package/shield, all screw-assembled.

- Iris placed λg/4 from the stripline's open-circuit end; strong magnetic field there excites the sphere's TM101-like hybrid mode.
- Via fence around the stripline suppresses surface-wave leakage.
- Efficiency measured with a mechanically height-tunable Wheeler cap (3–8 mm) to null internal cavity resonances before extracting radiation efficiency from the with/without-cap reflection difference.
- Results: S11 dip at 34.5 GHz (sim) / 35.0 GHz (meas) — mismatch blamed on K-connector contact and material-parameter approximation; ~3% impedance bandwidth (below benchmark, attributed to high εr = 10.25 → high Q → narrow BW); ~10 dBi peak gain, 11 dB directivity, ~74% simulated efficiency (≈ −1 dB); E-plane pattern shows ripples from microstrip/stripline transition leakage and multipath.
- Authors' own stated limitations: bandwidth needs improvement; the excited mode is a *hybrid*, not a pure natural Mie mode, because of feed/PCB/shield perturbation.

Paper 2 — Fundamental modes of a spherical DR at mmWave (MAPCON 2022)

- Establishes the physics (Mie theory, Q vs mode order, TE1q1 vs TM1q1 dominance conditions — see Part B) and validates it two ways: a WR-28 waveguide transmission test (TE1q1 zero at 28.5 GHz, TM1q1 zero at 33.1 GHz, fitted εr = 10.5, tan δ ≈ 10^{-3}) and a parallel-plate on-chip resonator prototype fed by microstrip fringing fields (TE1q1 at 33.02 GHz, Q≈2352; TM1q1 at 34.47 GHz, Q≈1363).
- Concludes TE1q1 is preferred for high-Q on-chip circuits (resonators, narrowband filters, sensors); TM1q1 for low-Q antenna/wideband applications.
- This paper is purely about single-element mode characterization — no radiation-focused antenna design, no multi-feed excitation, no polarization discussion.

The gap → your project's opening

Established by the two papers	Not yet addressed (your opportunity)
Single-sphere, single-feed excitation (stripline+iris in P1; microstrip fringe in P2)	Multi-feed excitation of one sphere for power combining
TM-mode antenna already demonstrated (P1)	A **TE-mode** radiating antenna has not been built/radiation-characterized in this pair
Self-alignment packaging solved	Feed-network design (splitters/phase shifters) for N feeds not addressed

Established by the two papers	Not yet addressed (your opportunity)
Linear/hybrid-dipole radiation only	No circular polarization treatment at all
Efficiency/Wheeler-cap methodology established	Needs re-validation once multiple feeds are combined

This table is effectively your "related work" section for the thesis introduction — cite both papers, then state the gap in one sentence: *neither work excites a spherical DR from multiple ports simultaneously, and neither addresses polarization control, which this project does by combining N feeds in a sequential-rotation arrangement to obtain both higher radiated power and circular polarization from a single TE1q1-mode sphere.*

Part D — Power Combining

Why combine power at mmWave

Solid-state sources (MMIC PAs) lose output power as frequency rises (device parasitics, lower breakdown voltage at smaller geometries), so a single amplifier/feed rarely delivers enough EIRP for a link budget. Two broad fixes: raise power per element (device-level, not an antenna problem) or combine several lower-power elements coherently (an antenna/feed-network problem — this is your project).

Circuit-level vs spatial combining

Approach	How it works	Loss behavior	Fit for this project
Circuit combining (Wilkinson, hybrid couplers, corporate feed trees)	Combine currents in a planar network before a single radiating port	Line/junction loss grows with number of ways and with feed-line length at mmWave	Needed as the feed network behind your multi-port sphere
Spatial / quasi-optical combining	Each source radiates (or excites) individually; combining happens in free space or in a shared resonant structure	Avoids lossy combiner lines at the final stage; loss shifts to phase/amplitude imbalance instead	This is effectively what a multi-feed DR does — the sphere itself is the free-space combining medium

Mapping this to a spherical DR

Your architecture is a hybrid: a planar N-way feed network (Wilkinson-tree or sequential rotation splitter) delivers correctly phased/amplitude-matched signals to N discrete probes/apertures arranged in a circle at the base of the sphere; the sphere then acts as the spatial combiner, superposing the N individual near-field excitations into one coherent TE1q1-like radiating mode.

Feed arrangement options (increasing complexity)

1. N probes/coax feeds in a circle, equal amplitude, equal phase — reinforces a single azimuthally-symmetric mode; simplest, but does not by itself give CP (see Part E for why phase must be staggered).
2. N probes, equal amplitude, phase stepped by 360°/N per feed (sequential rotation) — the classical trick from CP patch-array design; excites a rotating (circularly polarized) composite field pattern while simultaneously combining all N feed powers into one mode. This is your primary recommended architecture.
3. N aperture/stripline-iris feeds (Paper-1 style, scaled to N ports) — same idea, but coupling is via irises into striplines rather than direct probes; more design freedom in coupling strength (iris size) but a heavier multilayer PCB stack and more surface-wave/via-fence design work.
4. Hybrid: probes for initial proof-of-concept (fast to prototype, easy to phase individually with off-chip phase shifters/cables), then migrate to integrated aperture feeds once the concept is validated — a sensible thesis-timeline strategy.

Design variables that matter

- Number of feeds N (commonly 4 for a first CP design — gives 90° phase steps; can go higher for finer control or more combined power).
- Azimuthal spacing (uniform, 360°/N apart) and radius of the feed circle relative to the sphere's contact footprint.
- Amplitude balance across feeds (unequal amplitude degrades both combining efficiency and axial ratio).
- Feed-network loss budget: Wilkinson dividers are lossless in the ideal case only if perfectly matched/isolated; real insertion loss and phase/amplitude tracking error must be included in your CST/link budget.

What power combining efficiency means as a metric

Compare total radiated power (or peak realized gain) of the N-fed sphere against N times a single-feed element's radiated power. A ratio close to N (in linear power, or +10·log10(N) dB) indicates near-ideal coherent combining; any shortfall quantifies combining loss from phase/amplitude error and feed-network insertion loss.

Part E — Circular Polarization

Linear vs circular vs elliptical

A linearly polarized wave has E oscillating along a fixed line. A circularly polarized (CP) wave has two equal-amplitude, 90°-out-of-phase orthogonal field components, so the E-vector tip traces a circle in time at a fixed observation point — the polarization ellipse degenerates to a circle. Elliptical polarization is the general case in between. The standard figure of merit is axial ratio (AR), the ratio of the ellipse's major to minor axis, expressed in dB: AR = 0 dB is perfect CP, AR → infinity is linear. Practical CP antennas are specified by an AR bandwidth (commonly AR < 3 dB) which is usually the tightest bandwidth constraint in the whole design, tighter than the impedance (S11) bandwidth.

How CP is generated (three standard routes)

1. Dual-feed, 90° phase (a 90° hybrid/quadrature coupler feeding two orthogonal ports of one radiator) — simple, but only 2 feeds, limited combining benefit.
2. Single-feed perturbation (truncated corners on a patch, or an offset/perturbed feed on a single element) — no separate feed network needed, but narrow AR bandwidth and no power-combining angle at all.
3. Sequential rotation (N ≥ 3 elements or feeds, each physically rotated by 360°/N and fed with a matching 360°/N phase progression) — widely used in patch-array CP design specifically because it (a) produces good AR over a wider bandwidth than a single perturbed element, since amplitude/phase errors partially average out around the ring, and (b) naturally sums the power of all N feeds into one coherent output.

Why sequential rotation is the natural fit for this project

Sequential rotation is not really two separate techniques bolted together — feeding N ports around a circle with 360°/N phase steps *simultaneously* (a) spatially combines their power in the shared resonant structure (Part D) and (b) synthesizes CP, because the

phase-stepped, spatially-rotated excitation is mathematically a rotating field pattern by construction. This is why it is recommended as your primary architecture: one feed-network design choice buys both objectives of the project title.

Applying it to a TE1q1 sphere

The TE1q1 mode's field configuration (magnetic dipole, azimuthally circumferential H-field, see Part B and Fig. 4a of Paper 2) already has a natural axis of rotation. N feeds placed symmetrically around the sphere's equator/base, each phase-stepped by 360°/N, should excite a rotating version of this dipole field — the CP sense (RHCP/LHCP) is set by the direction of the phase progression around the ring. This is a hypothesis to validate in CST (Part G) — start with an eigenmode/field-symmetry check before committing to full 3D transient simulation with all N ports active.

What to actually verify in CST/measurement

- Axial ratio vs frequency along boresight (θ = 0°) and over the expected beamwidth.
- AR bandwidth vs the impedance-matching bandwidth (expect AR-BW to be the binding constraint).
- Sense of rotation (RHCP vs LHCP) matches your intended phase-progression direction.
- Degradation of AR with feed amplitude/phase imbalance (worth a parametric sweep — this quantifies fabrication sensitivity before you commit to hardware).

Part F — Proposed Project Architecture

Baseline concept

One alumina (or similar high-εr ceramic) sphere, self-aligned into an air-cavity in a supportive PCB layer exactly as in Paper 1, but fed by N ports (start with N = 4) arranged symmetrically in a circle around the sphere's base instead of a single stripline+iris. Primary feed style: direct coax/probe feeds into the substrate near the sphere's footprint (per your stated preference), with aperture/stripline-iris coupling kept as the documented alternative/comparison in your thesis (see comparison table below).

Feed network (behind the N ports)

An N-way Wilkinson-type power divider tree distributes the input signal equally, followed by a phase-shifting network (transmission-line length offsets, or lumped/printed phase shifters) that applies the 360°/N sequential-rotation phase steps before each probe. This network sits on the same or an adjoining PCB layer as the feed probes.

Stack-up sketch (build on Paper 1's layer logic)

1. Bottom: microstrip/coax launch + N-way divider + phase network.
2. Transition layer(s) to N vertical probes (or N stripline+iris ports if you pursue the aperture-coupled variant).
3. Supportive layer with the alignment air-hole, centered so all N feeds see the sphere symmetrically.
4. Sphere.
5. Metal shielding/package (as in Paper 1) sized so it has negligible interaction with the TE1q1 radiation, plus a low-permittivity foam/epoxy cover for mechanical stability.

Probe-fed vs aperture-coupled — comparison to include in your report

	Direct probe/coax feed (primary)	Aperture/stripline-iris feed (secondary, per Paper 1)
Design complexity	Lower — no iris/stripline-cavity design needed per port	Higher — each port needs its own iris + via-fenced stripline
Coupling control	Via probe position/depth, harder to fine-tune	Via iris size/position, more tunable coupling strength
Surface-wave risk	Lower (no long open stripline runs)	Higher, needs via-fencing (as in Paper 1)
Multilayer stack	Simpler, fewer layers	Heavier, more layers (as in Paper 1's exploded view)
Precedent in your reference papers	Not directly demonstrated (papers use stripline+iris)	Directly demonstrated for a single port
Scaling to N ports	Straightforward, N independent probes	More PCB real estate per added port

Recommendation matches your stated choice: build and validate with N direct probes first (faster CST iteration, fewer coupled EM effects to debug), then, time permitting, replicate one port as an aperture-coupled version for a controlled comparison in the thesis.

Material/substrate starting point (carried over from the two papers, safe defaults)

- Sphere: alumina, $\varepsilon_r \approx 10.2–10.5$, $\tan\delta \approx 0.005$ (Paper 1) or fitted $\approx 10^{-3}$ (Paper 2); 3 mm diameter is a proven, commercially available starting point (Goodfellow Cambridge Ltd. per Paper 2).
- Substrate: Rogers RO3003 ($\varepsilon_r = 3$, $\tan\delta = 0.001$) — same as both papers, well-characterized at Ka-band, good starting point for the feed network and supportive layer.
- Metallization: copper, ~17 μm, matching Paper 1's process.
- Target band: Ka-band (~28–40 GHz), consistent with both references and your sphere's TE1q1 resonance.

Open design question to settle before CST work starts

N = 4 is the standard sequential-rotation starting point (90° steps, manageable feed network), but the final N should trade off against: available PCB area at the sphere's base, feed-network loss (more ways = more divider stages = more loss), and desired combining gain ($+10\cdot\log_{10}(N)$ dB ideal). Worth an early back-of-envelope link-budget pass before committing to a value.

Part G — CST Studio Suite Workflow

Which solver, when

Solver	Use it for	Notes for this project
Eigenmode Solver	Finding a structure's natural resonant frequencies/modes with no feed/port	First step: characterize the bare sphere (in free space, then on your supportive layer, then near the metal shield) to confirm the TE1q1 frequency before adding any feed — mirrors Paper 2's parallel-plate PMC-symmetry approach
Time Domain (Transient) Solver	Broadband S-parameters, farfield, efficiency, in one wideband run; handles multiple ports well	Main workhorse for the fed, radiating structure — get S11..SNN and farfield/AR across the whole band in one solve
Frequency Domain Solver	High-accuracy narrowband results, resonant/high-Q structures, complex materials	Good cross-check near your TE1q1 resonance since it is a high-Q, narrowband feature that transient solvers can under-resolve if the mesh/timestep is coarse

Build → solve → evaluate flow

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flowchart TD
  A[Define units, frequency range, background material] --> B[Build geometry: sphere, substrate stack, feed probes/irises, shield]
  B --> C[Assign materials: eps_r, tan delta per layer + metal conductivity]
  C --> D[Set boundary conditions: open add space on radiating sides, PEC/PMC symmetry where valid]
  D --> E[Define ports: waveguide port or discrete port per feed]
  E --> F[Mesh: hex or tet, refine near sphere contact and feed gaps]
  F --> G[Eigenmode run: confirm bare-structure TE1q1 frequency]
  G --> H[Transient or Frequency Domain run: S-parameters, farfield, all ports active]
  H --> I[Post-process: S11..SNN, gain, directivity, efficiency, axial ratio]
  I --> J{Meets targets?}
  J -- no --> K[Parameter sweep / optimizer: iris size, probe position, phase steps]
  K --> F
  J -- yes --> L[Lock design, prepare for fabrication]
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Step notes

- Units and frequency range: set length in mm, frequency in GHz, and a range wide enough to see the full AR/impedance bandwidth plus some margin either side.
- Geometry: build layer-by-layer as in Paper 1′s exploded view (Fig. 1a) — supportive layer with the alignment hole, feed layer(s), shield. Use booleans to cut the air-hole and iris/via geometry cleanly.
- Materials: enter εr and tan δ per substrate/sphere layer, and real metal conductivity for copper/aluminum layers (not PEC) if you want realistic conductor-loss and efficiency numbers — both papers report exact σ values to use as a starting point.
- Boundary conditions: use open (add space) boundaries on the radiating side so the farfield is computed correctly; use PMC/PEC symmetry planes only where your geometry and excitation are truly symmetric (this halves/quarters simulation time) — remember Paper 2′s parallel-plate setup used a PMC symmetry plane specifically to isolate the sphere′s own Q from radiation.
- Ports: a coax/probe feed is usually a discrete port (or a coax waveguide port if you model the connector); an aperture/stripline feed is a waveguide port at the stripline cross-section. With N ports, CST computes the full N×N S-parameter matrix in one run — very useful for checking port-to-port isolation/coupling.

- Mesh: refine strongly near the sphere-substrate contact point and any narrow gaps (iris slot, probe-to-sphere gap) since these small features drive convergence; run a mesh-convergence check (increase mesh density until S11/gain stabilize) before trusting results.
- Eigenmode first: run the bare sphere (in free space, and separately seated on your actual supportive layer) with no ports to verify the TE1q1 frequency lines up with your target band before you spend time debugging a fed structure.
- Post-processing to extract: S11 (and full S-matrix for N ports), realized gain/directivity vs frequency, 3D and 2D (E-plane/H-plane) farfield patterns, radiation efficiency, and axial ratio vs frequency along boresight — CST has built-in axial ratio farfield plots once circular polarization components (LHCP/RHCP) are selected in the farfield monitor.
- Excitation phasing for N ports: CST lets you assign an amplitude and phase to each port's excitation signal (or you can combine individually-simulated single-port results with superposition in post-processing) — this is how you implement the sequential-rotation phase steps from Part E without needing N separate physical feed-network simulations at first.
- Parameter sweeps/optimizer: use CST's built-in parameter sweep (and optimizer for costlier searches) to tune iris/probe geometry, feed circle radius, and phase-step values against your S11/AR targets rather than hand-iterating.
- Macros (VBA/Python scripting in CST): useful once you are running the same sweep repeatedly (e.g. re-running the full N-port model for several sphere-tolerance values) — worth learning once your baseline single-feed model is working, not before.

Part H — Design Evaluation Checklist and Fabrication Path

Evaluation checklist (run every design iteration against this)

- Impedance match: S11 (and cross-port SNM for isolation) below −10 dB across the target band; report the −10 dB bandwidth as a percentage of center frequency, same convention as both papers.
- Gain / directivity / efficiency: peak realized gain along boresight, simulated radiation efficiency, and (once fabricated) measured efficiency via the Wheeler cap method from Paper 1.
- Axial ratio bandwidth: AR < 3 dB band, reported alongside impedance bandwidth — expect AR-BW to be the tighter constraint.

- Radiation pattern quality: check E-plane and H-plane cuts for symmetry and for ripples/spurious lobes (Paper 1 flagged E-plane ripples from feed-transition leakage — watch for the same with N ports).
- Power-combining efficiency: total radiated power/gain vs N times a single-port baseline (Part D).
- Mode purity: confirm (via eigenmode + field monitor plots) that the excited field still resembles the natural TE1q1 pattern and has not drifted into an uncontrolled hybrid mode from feed/shield perturbation (both papers flag this risk).
- CP sense correctness: RHCP vs LHCP matches the intended phase-progression direction.

From simulation to fabrication

- PCB stack-up and tolerances: follow Paper 1′s layer material choices (RO3003, defined copper thickness, via diameters/spacing) as a proven starting point; tighten tolerances where your N-port symmetry is sensitive to misalignment.
- Sphere sourcing: ceramic suppliers used in these papers (Goodfellow Cambridge Ltd., and similarly SITUS Technicals, Cospheric) supply alumina spheres in tightly toleranced diameters; order with sphericity/diameter tolerance specified, not just nominal size.
- Assembly: self-alignment via the air-hole in the supportive layer (as in both papers) plus alignment screws for layer registration; for N feeds, verify the sphere sits centered relative to all N ports simultaneously, not just one.
- Measurement setup: VNA with full N-port calibration (open-short-load-thru per port, as in Paper 2′s waveguide test), 2.92 mm or similar low-loss connectors, de-embedding of any connector/transition loss (Paper 1 explicitly separates out ~1 dB of stripline-to-microstrip transition loss).
- Efficiency measurement: mechanically height-tunable Wheeler cap (Paper 1′s method) to null internal cavity resonance before extracting radiation efficiency from with/without-cap reflection data.
- Pattern and AR measurement: anechoic chamber farfield measurement with a CP-capable measurement setup (rotating linear source or a dedicated CP antenna range) to extract axial ratio, not just gain.
- Expected sim-vs-measurement gaps to budget for: connector mechanical contact quality (both papers flag this), air gaps/misalignment between assembled layers, and dielectric parameters that are catalog/literature approximations rather than measured-in-house values — plan a small parameter-fitting step (as Paper 2 did) to reconcile simulated vs measured resonance frequency.

Part I — Timeline, Reading List, Glossary, Supervisor Q&A

Suggested milestone timeline

Phase	Milestone
1	Fundamentals solid (Parts A–B); replicate Paper 2's eigenmode/parallel-plate TE1q1 check in CST as a warm-up exercise
2	Single-probe-fed sphere: design, simulate, match to Ka-band, benchmark against Paper 1's numbers
3	Extend to N = 4 probes, equal phase: verify power-combining gain vs single element
4	Add sequential-rotation phasing: verify axial ratio, CP sense, re-check impedance match
5	Design the physical feed network (divider + phase shifters) to realize the phases used in simulation
6	Fabricate, assemble (self-alignment + screws), measure S-parameters, efficiency (Wheeler cap), pattern and AR
7	Reconcile sim vs measurement, iterate if needed, write up

Reading list

- Textbooks: Balanis, Antenna Theory: Analysis and Design (radiation, arrays, CP fundamentals); Pozar, Microwave Engineering (waveguide modes, Q-factor, S-parameters); Bohren & Huffman, Absorption and Scattering of Light by Small Particles (Mie theory, cited as [15] in Paper 2); Kajfez & Guillon, Dielectric Resonators (general DR theory).
- Follow-on papers to chase from the two papers' own reference lists: Dey & Hesselbarth on self-aligned on-chip spherical resonators and 180 GHz spherical DRA [7],[8] in Paper 1; Sterzl et al. on self-aligned on-chip resonators [7] in Paper 2; any CP-

DRA or sequential-rotation array papers (search terms: sequential rotation circularly polarized array, circularly polarized dielectric resonator antenna, spatial power combining antenna array).

Glossary (quick lookup)

- TEM / TE / TM: see Part A table.
- Q-factor: energy stored / energy dissipated per cycle, times omega-naught.
- Size parameter x: 2·pi·r/lambda, governs which Mie mode dominates.
- Mie coefficients a_n, b_n, c_n, d_n: external TM, external TE, internal TE, internal TM mode amplitudes of order n.
- Axial ratio (AR): ratio of major to minor axis of the polarization ellipse; 0 dB = perfect CP.
- Sequential rotation: feeding N elements with phase stepped by 360°/N around a physical rotation of 360°/N, used for CP and power combining together.
- Wheeler cap: a shielded cavity method to extract radiation efficiency from reflection-coefficient measurements with and without the cap.

Quick-reference Q&A for supervisor meetings

- What is a transverse wave? A wave whose field vectors oscillate perpendicular to its direction of travel.
- How do they travel? As oscillating E and H fields satisfying Maxwell's equations, propagating with phase constant beta along the guide/space, at a speed set by the medium (c/sqrt(eps_r) in a simple dielectric).
- What is TEM mode? A mode with no longitudinal E or H component at all; only exists where two separate conductors are present to support the transverse static-like field pattern.
- TE and TM mode profile? TE has no longitudinal E (Ez = 0, Hz not equal to 0); TM has no longitudinal H (Hz = 0, Ez not equal to 0); the profile is the fixed transverse field shape across the cross-section set by geometry and boundary conditions.
- What profile do we generally deal with? Quasi-TEM on the feedlines, TE10 if characterizing via WR-28 waveguide, and TE1q1/TM1q1 Mie modes inside the sphere itself — this project specifically targets TE1q1.
- What is a resonator? A structure that stores oscillating field energy at discrete natural frequencies set by its boundary conditions.

- Why a dielectric sphere specifically? No conductor loss, single tolerance dimension, smallest planar footprint, good commercial availability at tight tolerance.
- How do you combine power in this design? Multiple feeds around the sphere, phase-stepped for sequential rotation, so the sphere spatially combines them into one coherent radiating mode while simultaneously producing circular polarization.

All of the above traces directly back to Parts A through H above should a follow-up question need the fuller derivation or the specific paper citation.